Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected tree-fitting record.

Name fitted parts, audit both candidates, preserve validation evidence and explain the tie rule.

Keep one habit: Retain training counts, candidate scores, saved leaves, predictions and limitations.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Types of Classification Error

Training and Testing

Operational Labels

W1 · Name the learned root and leaves

> Distinguish chosen class names from fitted predictions.

Your evidence record

- Fitting policy: candidates near first, tall second; flags use0/1. For each candidate and each branch count only its training labels, predict that branch’s majority and choose clear for a tied leaf count. Add all training mistakes, replace the best candidate only for strictly fewer errors and stop at depth one. Freeze the selected question and both leaves for validation. Choose the baseline from the training-majority label, with clear for a tie. Worked training rows (near,tall,label), IDs1–6: (1,0,blocked),(1,1,blocked),(1,0,blocked),(0,1,clear),(0,0,clear),(0,1,blocked). Worked validation rows: (1,1,blocked),(0,0,clear),(0,1,blocked),(1,0,clear). The independent W4 supplies its own different rows.

- Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Count each candidate’s branch labels

> Sum row mistakes rather than averaging branch sizes.

Your evidence record

- Fitting policy: candidates near first, tall second; flags use0/1. For each candidate and each branch count only its training labels, predict that branch's majority and choose clear for a tied leaf count. Add all training mistakes, replace the best candidate only for strictly fewer errors and stop at depth one. Freeze the selected question and both leaves for validation. Choose the baseline from the training-majority label, with clear for a tie. Worked training rows (near,tall,label), IDs1–6: (1,0,blocked),(1,1,blocked),(1,0,blocked),(0,1,clear),(0,0,clear),(0,1,blocked). Worked validation rows: (1,1,blocked),(0,0,clear),(0,1,blocked),(1,0,clear). The independent W4 supplies its own different rows.

- For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Preserve the original fitted tree during scoring

> Record any validation-guided revision as a later version.

Your evidence record

- Fitting policy: candidates near first, tall second; flags use0/1. For each candidate and each branch count only its training labels, predict that branch's majority and choose clear for a tied leaf count. Add all training mistakes, replace the best candidate only for strictly fewer errors and stop at depth one. Freeze the selected question and both leaves for validation. Choose the baseline from the training-majority label, with clear for a tie. Worked training rows (near,tall,label), IDs1–6: (1,0,blocked),(1,1,blocked),(1,0,blocked),(0,1,clear),(0,0,clear),(0,1,blocked). Worked validation rows: (1,1,blocked),(0,0,clear),(0,1,blocked),(1,0,clear). The independent W4 supplies its own different rows.

- A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Explain the candidate tie convention

> Candidate order is a procedure choice, not evidence of future superiority.

Your evidence record

- Fitting policy: candidates near first, tall second; flags use0/1. For each candidate and each branch count only its training labels, predict that branch's majority and choose clear for a tied leaf count. Add all training mistakes, replace the best candidate only for strictly fewer errors and stop at depth one. Freeze the selected question and both leaves for validation. Choose the baseline from the training-majority label, with clear for a tie. Worked training rows (near,tall,label), IDs1–6: (1,0,blocked),(1,1,blocked),(1,0,blocked),(0,1,clear),(0,0,clear),(0,1,blocked). Worked validation rows: (1,1,blocked),(0,0,clear),(0,1,blocked),(1,0,clear). The independent W4 supplies its own different rows.

- If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Fit a new tree independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Fitting policy: candidates near first, tall second; flags use0/1. For each candidate and each branch count only its training labels, predict that branch's majority and choose clear for a tied leaf count. Add all training mistakes, replace the best candidate only for strictly fewer errors and stop at depth one. Freeze the selected question and both leaves for validation. Choose the baseline from the training-majority label, with clear for a tie.

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Fit a new tree independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Create a transparent tree for hard constraints before applying preference scores.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Verify the candidate scores for **Fitting** a **decision tree**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Fitting** a **decision tree**. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A tree asks topic is space, then pages at most 60. A space book has 70 pages. Trace both answers and explain why you cannot stop at the first question.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 14 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

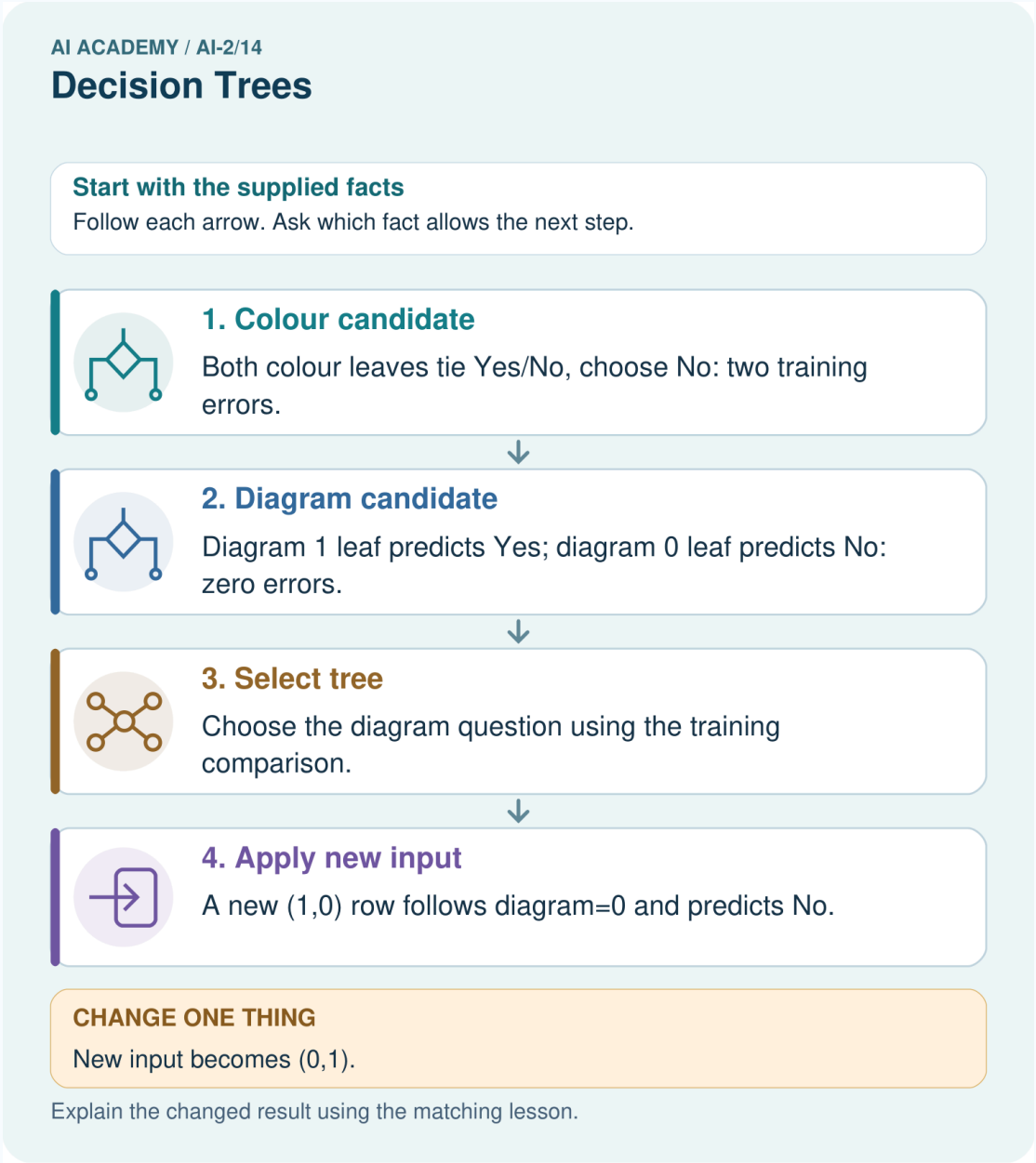
> Keep the explanation and the support you used.

Use this when helpful

A fictional toy classifier predicts whether a book needs an illustration label.

Training rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why does changing colour alone not change this tree's **prediction**?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy classifier predicts whether a book needs an illustration label.
- Which labelled step matches this detail? A new (1,0) row follows diagram=0 and predicts No.
- 1. Select tree
- 2. Diagram candidate
- 3. Colour candidate

- 4. Apply new **input**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Apply new **input** A new (1,0) row follows diagram=0 and predicts No. A question becomes a fitted tree only after its leaves and selection procedure are specified.

The **training** comparison prefers the feature that separates these labels. **Prediction** follows the selected question; it does not reselect the tree using a new case's answer.

Save candidate leaves and errors so the choice can be repeated.

Changed-case question

- Changed-case practice: New **input** becomes (0,1). Which conclusion is supported by the supplied facts?

- 1. The fitted tree predicts Yes.

- 2. This one example guarantees correct results for every future **input**.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The fitted tree predicts Yes. It follows diagram=1; cover colour is not the selected question.

Repair and save your work

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Decision Trees: the case

AI ACADEMY / AI-2/14 · 01 · LOOK AT THE FACTS

Decision Trees: the case

The situation

A fictional toy classifier predicts whether a book needs an illustration label.

Supplied evidence

Training rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No). Candidates ask colour_cover or diagram_inside; each leaf uses its training majority. A leaf tie chooses No. Select fewer training errors.

By the end, I can fit a one-question tree using training labels and an explicit candidate comparison.

Follow the labelled picture with the matching explanation. Open the vector diagram

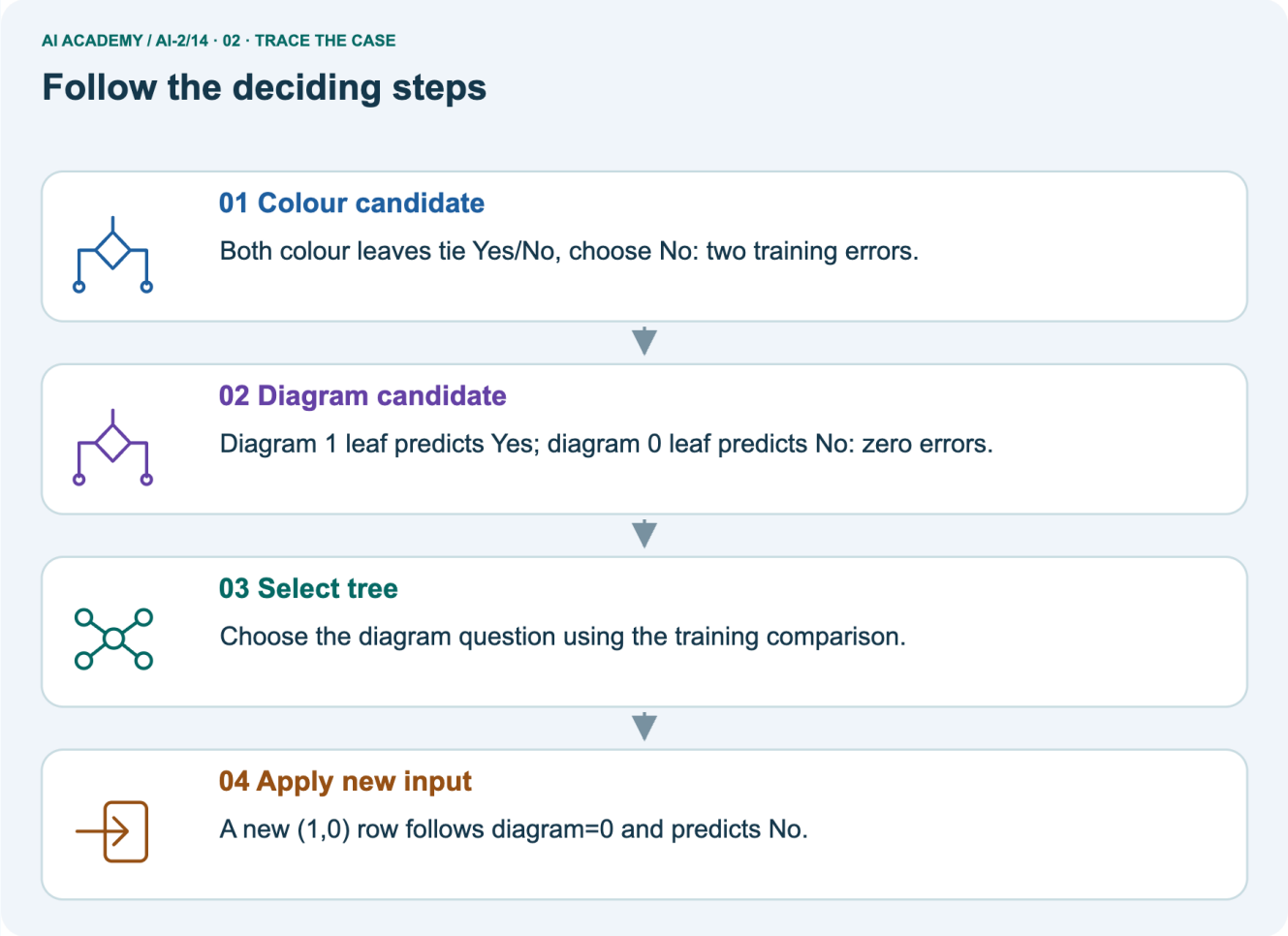
A fictional toy classifier predicts whether a book needs an illustration label. Supplied facts: **Training** rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Colour candidate: Both colour leaves tie Yes/No, choose No: two **training** errors. → Diagram candidate: Diagram 1 **leaf** predicts Yes; diagram 0 **leaf** predicts No: zero errors. → Select tree: Choose the diagram question using the **training** comparison. → Apply new **input**: A new (1,0) row follows diagram=0 and predicts No.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two paths through the learned tree

AI ACADEMY / AI-2/14 · 03 · INSPECT THE LESSON MODEL

Two paths through the learned tree

Root question	Branch answer	Learned leaf
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

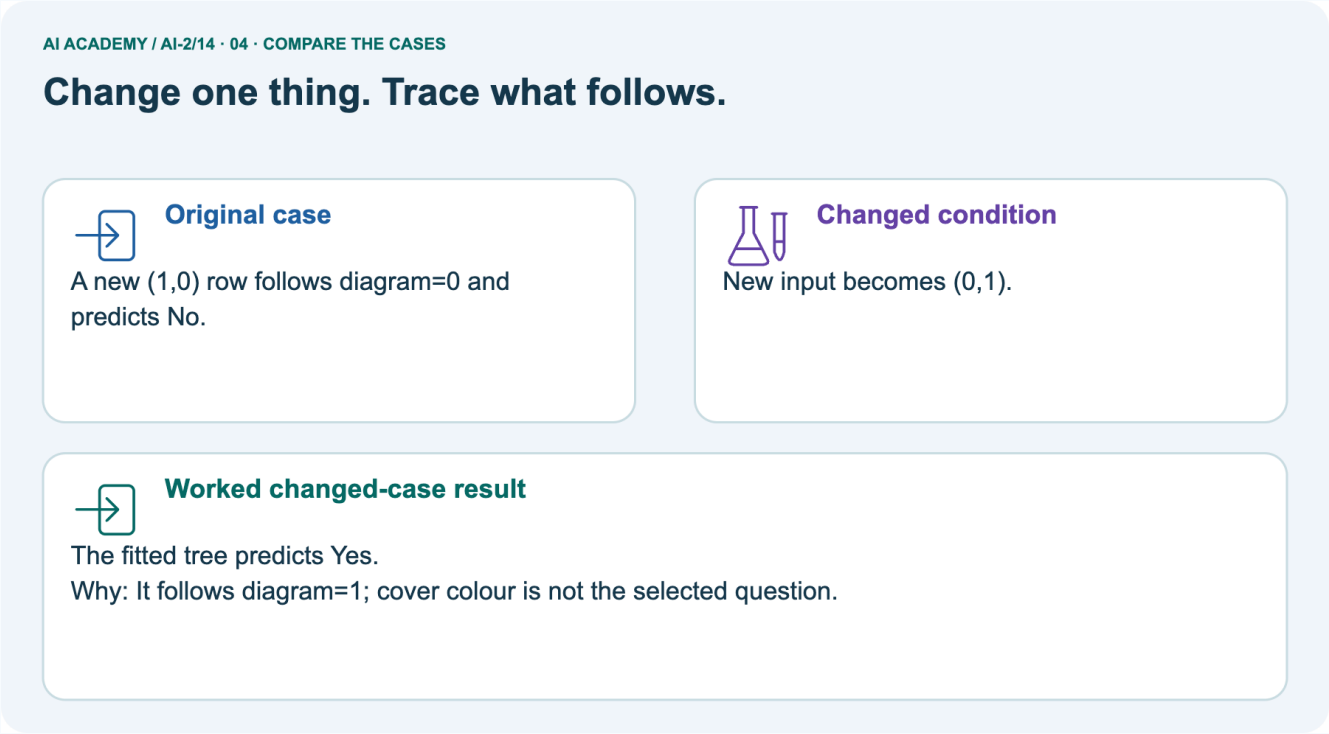
Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

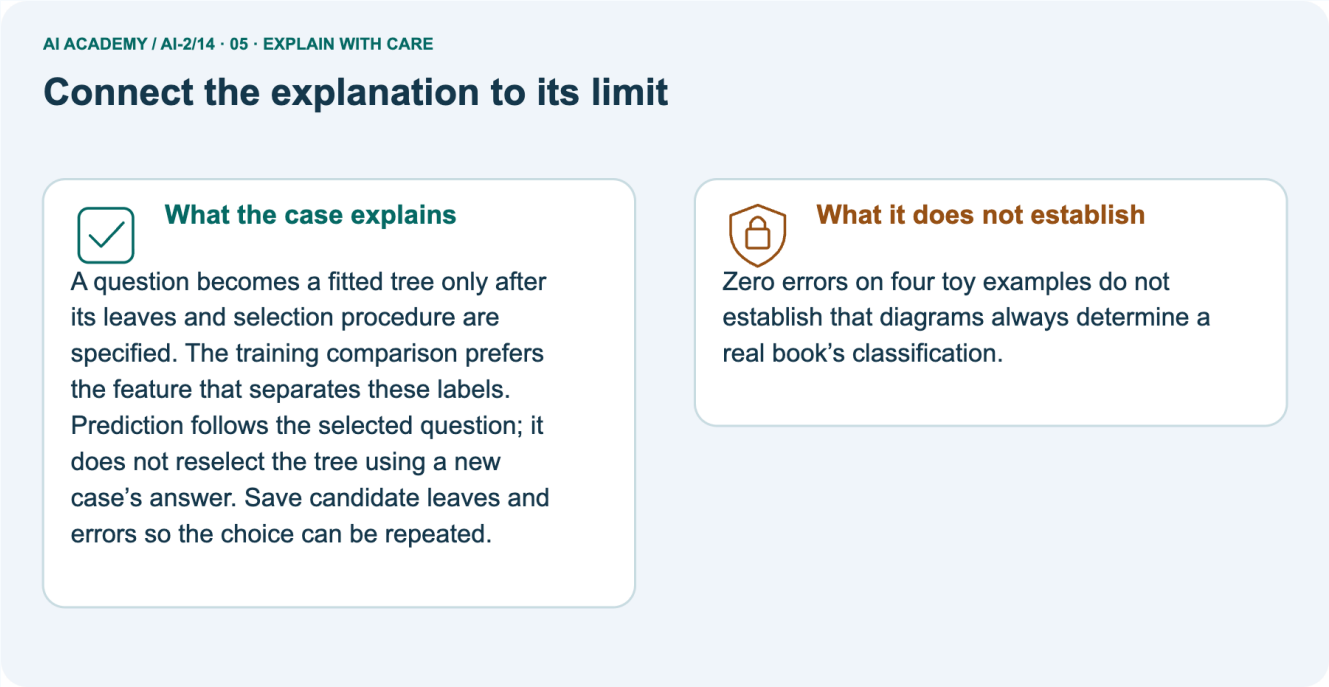
Original case: A new (1,0) row follows diagram=0 and predicts No. Changed condition: New **input** becomes (0,1). Expected result: The fitted tree predicts Yes.

Reason: It follows diagram=1; cover colour is not the selected question.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

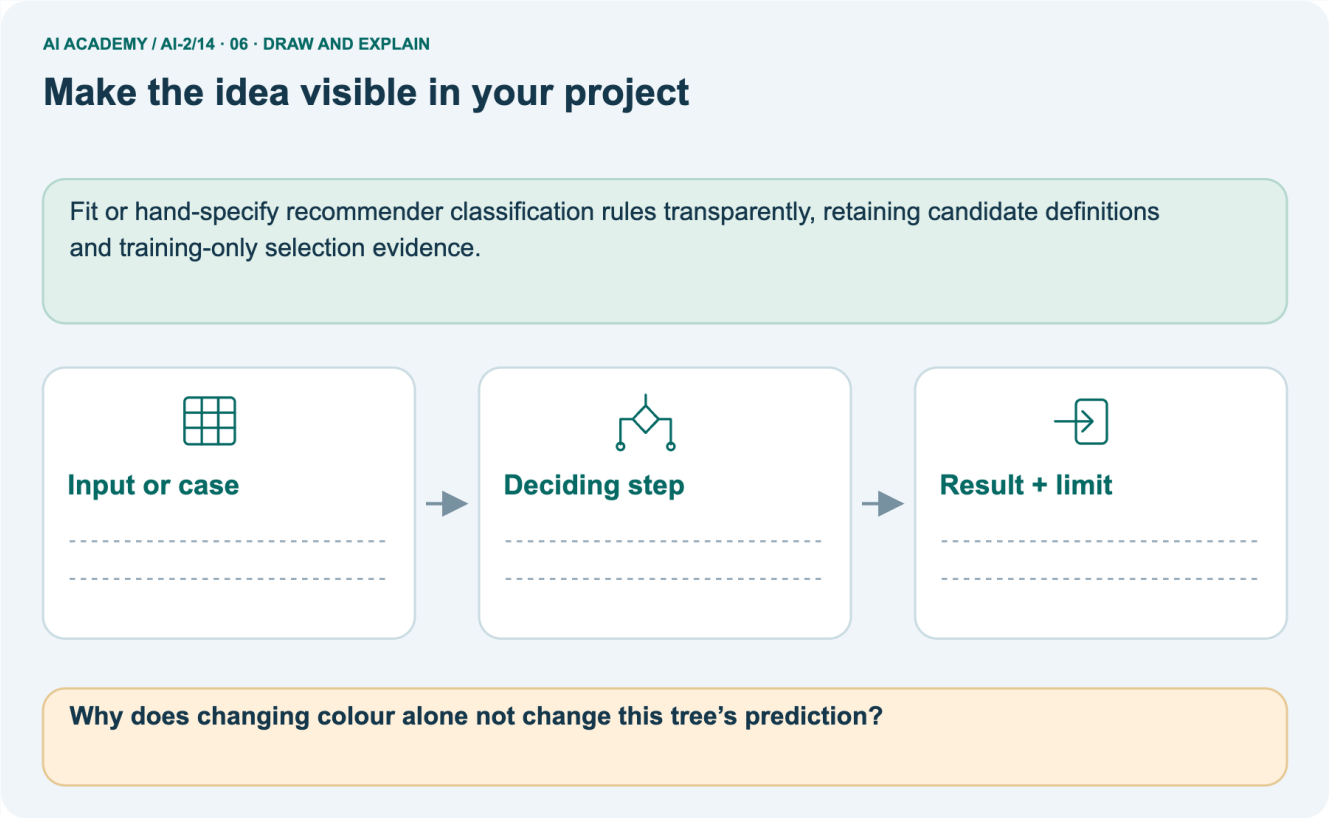
Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Limit: Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Why does changing colour alone not change this tree's **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Decision Trees: the case

02. Follow the deciding steps

03. Two paths through the learned tree

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.